

[25-26] Computer Vision and Image Processing [MEI]

English translation of the provided course page and assessment brief

Course Status

Open

Course Contents

Lecture 09 - Peeking inside CNNs

Lecture 08 - UNets, GANs

Lecture 07 - Autoencoders

Lecture 6 - Ensembles; Transfer Learning; ROC

Lecture 5 - Data Augmentation

Lecture 04 - Layers and Validation Set

Lecture 03 - Fully Connected vs. CNN

Lecture 02 - MNIST with a feed-forward model

Assessment

Software required to run locally

Module Content

Individual Assignment - Due date: 20/06/26, 00:59 (CEST)

Group Assignment - Phase 1 - Due date: 17/05/26, 00:59 (CEST)

Group Assignment - Phase 2 - Due date: 06/06/26, 00:59 (CEST)

Support material related to the Image Processing modules taught during the first 4 weeks

DUC LTI

Teaching Staff

Name	Role
António José Borba Ramires Fernandes	Instructor
Luís Paulo Peixoto Santos	Instructor

Details and Actions

List - View all participants in your course

Course description - View the course description

Class Collaborate - Join the session

Manuals and tools - View the tools for your course and institution

Assessment

This course unit will be assessed based on two practical assignments: the first to be completed in a group, and the second individually.

Assignment 1 Brief - Group (2-3 members)

This assignment aims to explore deep learning models applied to the GTSRB dataset (German traffic sign dataset) in order to obtain the best possible result in terms of accuracy on the test dataset. The best published result is 99.82%.

The assignment is divided into two parts:

In the first part, students are expected to train models using data augmentation, both as preprocessing and dynamically. In this part, students should explore image processing filters and methods and evaluate their impact on the final performance of the network. Part of the assessment will be determined by the image processing techniques used. Several models with different types of data augmentation should be trained, and an analysis of the obtained results should be carried out.

In the second phase, students are expected to study the potential of using ensembles of networks. The networks included in the ensemble will be those trained in the first phase.

German Dataset

Training images (ZIP)

Test images (ZIP)

In addition to the notebook (or standalone Python scripts), students must submit a short report describing the choices made and presenting the tests carried out. The notebook or scripts must also be properly commented.

Submission via Blackboard.

The assignments will have a public presentation on June 6. Groups should prepare a PowerPoint for the presentation.

Final submission dates: Phase 1: May 16; Phase 2: June 5; Presentation on June 5.

Assignment 2 Brief - Individual

This assignment asks students to explore GANs and/or Autoencoders in a problem of their choice. There are many datasets available for training this type of network.

The main objective is to explore loss functions, for example in line with what was done in Pix2Pix or CycleGAN, and/or to explore manipulations and combinations of datasets.

The report should be concise and may be included in the notebook itself. It should contain a description of the strategy defined by the students, as well as a discussion/comparison of the results obtained. In the case of comparison, this should be based on results published by third parties.

Submission via Blackboard.

Final submission date: June 19.

Deadlines Mentioned on the Course Page

Item	Date / Time
Individual Assignment	20/06/26, 00:59 (CEST)
Group Assignment - Phase 1	17/05/26, 00:59 (CEST)
Group Assignment - Phase 2	06/06/26, 00:59 (CEST)
Assignment 1 final submission - Phase 1	May 16
Assignment 1 final submission - Phase 2	June 5

Assignment 1 presentation	June 5 / also mentioned as June 6 in the brief
Assignment 2 final submission	June 19

Note: The provided text contains slightly different date references between the course content list and the assessment brief. This translation preserves those date references as shown.